

Report on Text Summarization Using Transformers

Background

This report summarizes the utilization of the BART model (Bidirectional and Auto-Regressive Transformers) model from the transformer's library. The task involved summarizing chapters from Jane Austen's classic novel "Pride and Prejudice," obtained from Project Gutenberg. The primary objective was to evaluate different hyperparameter configurations to identify the optimal settings for generating high-quality summaries that balance conciseness, informativeness, and readability.

Transformer Architecture and Implementation

BART Model Overview

The BART model was selected for this summarization task. BART is a sequence-to-sequence denoising autoencoder pre-trained on a large corpus of English text. The model combines:

- A bidirectional encoder similar to BERT that processes the input text
- An auto-regressive decoder similar to GPT that generates output text
- Attention mechanisms that allow the model to focus on relevant parts of the input when generating each word of the summary

The specific pre-trained model used was facebook/bart-large-cnn. This model contains 406 million parameters and is optimized for generating concise, informative summaries.

Implementation Details

The implementation utilized the transformers library from Hugging Face, which provides pre-trained transformer models and tokenizers. The model was loaded with BartForConditionalGeneration, and text was tokenized using BartTokenizer.

Hyperparameter Tuning

Four distinct hyperparameter configurations were evaluated to identify optimal settings for summary generation. Each configuration was tested on five sample chapters, and summaries were evaluated using ROUGE metrics against extractive reference summaries.

Configuration	max_length	min_length	num_beams	length_penalty	no_repeat_ngram
Config 1: Balanced	150	50	4	2.0	3

Configurati on	max_leng th	min_lengt h	num_bea ms	length_penal ty	no_repeat_ngr am
Config 2: Longer Summary	200	80	4	2.0	3
Config 3: Concise	100	30	4	2.5	3
Config 4: High Quality	150	50	8	2.0	3

Hyperparameter Explanation

- **max_length:** Maximum number of tokens in the generated summary.
- **min_length:** Minimum number of tokens required in the summary.
- **num_beams:** Number of beams for beam search. Higher values explore more potential sequences, improving quality but increasing computation time.
- **length_penalty:** Exponential penalty applied to sequence length. Values >1.0 encourage longer sequences, while values <1.0 favor shorter outputs.
- **no_repeat_ngram_size:** Size of n-grams that cannot be repeated. This prevents the model from generating repetitive phrases.

Evaluation Methodology

Model performance was evaluated using ROUGE (Recall-Oriented Understudy for Gisting Evaluation) metrics, which are standard for assessing summarization quality. Since the task involved generating summaries without pre-existing references, extractive reference summaries were created by selecting representative sentences from each chapter (first sentence, last sentence, and middle sentences).

ROUGE Metrics:

- **ROUGE-1:** Measures overlap of unigrams (individual words) between the generated summary and reference summary. Indicates vocabulary coverage.
- **ROUGE-2:** Measures overlap of bigrams (two consecutive words). Captures phrase-level similarity.
- **ROUGE-L:** Based on longest common subsequence, measuring sentence-level structure similarity. Accounts for word order and sentence structure.

Additionally, compression ratio (original length / summary length) was calculated to quantify the degree of information condensation achieved by each configuration.

Results and Analysis

Configuration	ROUGE-1	ROUGE-2	ROUGE-L	Compression	Avg Length
Config 1: Balanced	0.2227	0.0065	0.1225	28.45	49.6
Config 2: Longer Summary	0.2456	0.0248	0.1458	18.42	74.0
Config 3: Concise	0.1952	0.0027	0.1121	34.71	41.0
Config 4: High Quality	0.2053	0.0098	0.1192	28.00	51.2

Key Findings

- Best Overall Performance:** Configuration 2 (Longer Summary) achieved the highest ROUGE scores across all metrics, with:
 - ROUGE-1 of 0.2456
 - ROUGE-2 of 0.0248
 - ROUGE-L of 0.1458.
 This configuration also provided the best balance between information retention and conciseness with a compression ratio of 18.42.
- Length vs. Quality Trade-off:** Longer summaries (Config 2: 74 words average) captured more information from the source text compared to concise summaries (Config 3: 41 words average), as evidenced by higher ROUGE scores. The substantial difference in ROUGE-2 scores (0.0248 vs. 0.0027) indicates that longer summaries maintained better phrase-level coherence.
- Consistency Analysis:** Configuration 2 demonstrated relatively low standard deviations (ROUGE-1: ± 0.0492 , ROUGE-2: ± 0.0099 , ROUGE-L: ± 0.0328), indicating consistent performance across different chapters with varying content and structure.

Conclusion

This analysis demonstrates that BART is capable of producing high-quality summaries of literary text. Among the tested configurations, Configuration 2 (Longer Summary) yielded the best results, with parameters set to `max_length=200`, `min_length=80`, `num_beams=4`, `length_penalty=2.0`, and `no_repeat_ngram_size=3`. Longer summaries effectively retain more content while maintaining conciseness.

The relatively low ROUGE scores can be attributed to the metric's reliance on exact lexical overlap between generated and reference summaries. BART generates abstractive summaries, rephrasing ideas rather than copying sentences verbatim. In contrast, the reference summaries were extractive, constructed directly from

sentences in the source text. This discrepancy reduces word-level overlap, causing ROUGE to underestimate the actual semantic quality of the generated summaries.